Machine Learning

Haystack Dense Retrieval Practice Document

Introduction

Machine Learning (ML) is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every situation. ML systems improve their performance when they receive more useful data and feedback.

Types of Machine Learning

The main types are supervised learning, unsupervised learning, and reinforcement learning. Supervised learning uses labelled examples for tasks such as classification and regression. Unsupervised learning finds patterns in data without labels, including clustering. Reinforcement learning learns actions through rewards and penalties.

Common Algorithms

Common machine learning algorithms include linear regression, logistic regression, decision trees, random forests, support vector machines, k-nearest neighbors, and neural networks. The choice of algorithm depends on the data, problem, and performance requirements.

Applications

Machine learning is used in spam detection, recommendation systems, fraud detection, medical image analysis, demand forecasting, customer segmentation, speech recognition, and computer vision.

Benefits and Limitations

ML can automate repetitive decisions, discover patterns, and support data-driven predictions. Limitations include poor data quality, bias, overfitting, high computational requirements, and difficulty explaining some complex models.

This document is prepared as a small knowledge source for a Haystack BM25 versus dense retrieval experiment.